

Causal identification in macro IV

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Sign restrictions

- Much of SVAR/SVMA inference relies on exclusion restrictions, i.e., certain functions of parameters are assumed to equal zero.
- Except possibly in the case of a credible proxy/IV, exact zero restrictions may not be so plausible.
- Sometimes economic theory/intuition tells us that the impulse responses of certain observed variables to certain shocks should have a specific *sign*.
 - ▶ Output responds negatively to a contractionary monetary policy shock.
 - ▶ Lending contracts and spreads rise following a bad financial shock.
- How can we do SVAR/SVMA inference and only impose these **sign restrictions**?

Sign restrictions: SVAR model

- Although sign restrictions could in principle be imposed in the SVMA model, historically they have been used mainly in conjunction with the invertibility assumption.

- Assume an SVAR model:

$$A(L)y_t = \eta_t, \quad \eta_t = H\varepsilon_t, \quad \varepsilon_t \sim WN(0, I_n).$$

- Reduced-form impulse responses

$$B(L) = I_n + \sum_{\ell=1}^{\infty} B_{\ell}L^{\ell} \equiv A(L)^{-1}.$$

- Structural impulse responses

$$\Theta(L) = \sum_{\ell=0}^{\infty} \Theta_{\ell}L^{\ell} \equiv B(L)H = B(L)CQ,$$

where $CC' = \Sigma \equiv \text{Var}(\eta_t)$ (e.g., Cholesky) and $QQ' = I_n$.

- $A(L)$, $B(L)$, and C are identified, but Q is totally unknown.

Sign restrictions: identified set for Q

- Suppose we want to impose the restriction that $\Theta_{i,j,\ell} \geq 0$ for some (i,j,ℓ) . This amounts to

$$B_{i,\bullet,\ell} C Q_{\bullet,j} \geq 0. \quad (\dagger)$$

- This is an inequality restriction on a linear combination of a column of Q . Remember, the vector $B_{i,\bullet,\ell} C$ is point-identified.
- The **identified set** for Q are all those values of Q that satisfy the sign restrictions of the form $(\dagger)$ that we want to impose, as well as the $\frac{n(n+1)}{2}$ non-redundant quadratic constraints in the equation

$$Q Q' = I_n. \quad (\Delta)$$

- The identified set contains all the values of Q that are jointly consistent with (i) the data and (ii) our sign restrictions.

Application: fiscal shocks

- Mountford & Uhlig (2009) identify the responses of U.S. economic variables to various shocks, including fiscal shocks.
- They impose the following sign restrictions on impulse responses. The restrictions are imposed at horizons $\ell = 0, 1, 2, 3, 4$ (quarters).

Table I. Identifying sign restrictions

	Gov. revenue	Gov. spending	GDP, cons, non-res.inv.	Interest rate	Adjusted reserves	Prices
<i>Non-fiscal shocks</i>						
Business cycle	+		+			
Monetary policy				+	-	-
<i>Basic fiscal policy shocks</i>						
Government revenue	+					
Government spending		+				

This table shows the sign restrictions on the impulse responses for each identified shock. 'Cons' stands for private consumption and 'Non-res. inv.' stands for non-residential investment. A '+' means that the impulse response of the variable in question is restricted to be positive for four quarters following the shock, including the quarter of impact. Likewise, a '-' indicates a negative response. A blank entry indicates that no restrictions have been imposed.

Sign restrictions: partial identification

- For given $B(L)$ and C , the identified set for Q may be...
 - ▶ Empty (the restrictions are inconsistent with the ACF of the data).
 - ▶ A singleton (happens only in knife-edge cases).
 - ▶ A set with multiple elements, usually uncountably many.
- The last case is the typical one. In this case, we say that the model is **partially identified** (a.k.a. **set-identified**).
- In the partially identified case, there are (often infinitely) many different parameter values that are consistent with the data and our restrictions, even in an infinitely large data sample.

Sign restrictions: identified set for IRFs

- We don't care about Q *per se*. Suppose we want **upper bound** and **lower bound** on an impulse response $\Theta_{i^*, j^*, \ell^*}$ for some (i^*, j^*, ℓ^*) .
- We then solve the optimization program

$$\sup_Q / \inf_Q \quad B_{i^*, \bullet, \ell^*} C Q_{\bullet, j^*}$$

subject to the linear inequality constraints of the form $(\dagger)$ as well as the quadratic equality constraints (Δ) .

- This is a quadratic programming problem subject to inequality constraints. Fairly well-behaved computational procedures exist (Gafarov, Meier & Montiel Olea, 2018; Giacomini & Kitagawa, 2021).
- Less fancy approach: Approximate min/max by repeatedly simulating random orthogonal Q 's (more later).
- Could similarly get bounds on FVDs (Faust, 1998; Volpicella, 2022).

Analytical example: bivariate i.i.d. model

$$\begin{pmatrix} y_{1,t} \\ y_{2,t} \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}, \quad \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} \overset{i.i.d.}{\sim} N(0_{2 \times 1}, I_2).$$

- We normalize $H_{11}, H_{22} \geq 0$ (defines signs of the shocks).
- In addition we impose the substantive sign restriction $H_{21} \geq 0$.
- What is the *largest* value of H_{12} consistent with $\Sigma \equiv \text{Var}(y_t)$?
- Define

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{21} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \quad C = \begin{pmatrix} C_{11} & 0 \\ C_{21} & C_{22} \end{pmatrix} \equiv \begin{pmatrix} \sqrt{\Sigma_{11}} & 0 \\ \frac{\Sigma_{21}}{\sqrt{\Sigma_{11}}} & \sqrt{\Sigma_{22} - \frac{\Sigma_{21}^2}{\Sigma_{11}}} \end{pmatrix}$$

so $CC' = \Sigma$. The diagonal elements of C are positive.

Analytical example: bivariate i.i.d. model (cont.)

- We have

$$H = CQ = \begin{pmatrix} C_{11}Q_{11} & C_{11}Q_{12} \\ C_{21}Q_{11} + C_{22}Q_{21} & C_{21}Q_{12} + C_{22}Q_{22} \end{pmatrix},$$

where $QQ' = I_n$. Restriction $H_{11} \geq 0$ gives $Q_{11} \geq 0$.

- Note that $H_{12} \leq C_{11}$, since $|Q_{12}| \leq \|Q_{\bullet,2}\| = 1$.
- Case $\Sigma_{21} \geq 0$: Then $C_{21} \geq 0$. The largest value in the identified set for H_{12} is $C_{11} = \sqrt{\Sigma_{11}}$. This is achieved by $Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which satisfies all three sign restrictions.
- Case $\Sigma_{21} < 0$: **Exercise.** Hint: Orthogonality of Q implies that either $Q = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ or $Q = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$ for some $\theta \in [0, 2\pi)$.

Sign restrictions and zero restrictions

- It's easy to additionally impose hard zero restrictions on impulse responses.
- The assumption $\Theta_{i,j,\ell} = 0$ places the extra linear restriction

$$B_{i,\bullet,\ell} C Q_{\bullet,j} = 0$$

on Q .

- We can then do our optimization over Q imposing both inequality and equality restrictions.
- Zero restrictions (if they are credible) can be very useful for narrowing down the identified set.

Sign restrictions: full Bayesian inference

- It's not so easy to do frequentist inference in partially identified settings, since we don't have consistent estimators of all parameters (but more on this later).
- However, in principle, Bayesian inference is straight-forward.
 - ▶ Choose a prior for all parameters.
 - ▶ Combine with model-implied likelihood function through Bayes' rule.
 - ▶ MCMC using posterior density.
- Nothing in this procedure depends explicitly on whether we are point-identified or not.
- A committed subjective Bayesian need not worry about partial ID. Not true if there's some ambiguity about prior (more later).

Sign restrictions: full Bayesian inference (cont.)

- It is common in the literature to adopt the conjugate normal-inverse-Wishart prior for the reduced-form VAR parameters $(A(L), \Sigma)$. Computationally convenient.
- We're not done: We still need a prior for Q , and it has to satisfy the sign restrictions we impose.
- Q lives in the space $\mathbb{S}_n$ of orthogonal $n \times n$ matrices.
- One famous probability distribution on $\mathbb{S}_n$ is **Haar measure**. We can simulate from it as follows:
 - ▶ Draw a $n \times n$ matrix X whose entries are i.i.d. $N(0, 1)$.
 - ▶ Let $X = QR$ be the **QR decomposition** of X (Matlab command: `qr`). The upper triangular matrix R is normalized to have positive diagonal.
 - ▶ Then Q is a draw from Haar measure.
- Haar measure is sometimes called the “uniform distribution” on $\mathbb{S}_n$. This is just a name.

Sign restrictions: full Bayesian inference (cont.)

- As a conditional prior for Q given $(A(L), \Sigma)$, Uhlig (2005) and Rubio-Ramírez, Waggoner & Zha (2010) suggested using Haar measure *truncated* to the space where the sign restrictions hold.
- Assume the shocks ε_t are i.i.d. Gaussian:

$$A(L)y_t = C(\Sigma)Q\varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0_{n \times 1}, I_n).$$

Second moments fully determine the likelihood function.

- Thus, Q does not affect the likelihood function at all, as long as parameters satisfy the sign restrictions:

$$p(\mathcal{Y}_T \mid A(L), \Sigma, Q) = p(\mathcal{Y}_T \mid A(L), \Sigma, \tilde{Q})$$

for any $Q, \tilde{Q} \in \mathbb{S}_n$ such that the sign restrictions hold given $(A(L), \Sigma)$.

- Thus, we do not do Bayesian updating about Q at all, except we rule out any triplet $(A(L), \Sigma, Q)$ which violates the sign restrictions.

Sign restrictions: full Bayesian inference (cont.)

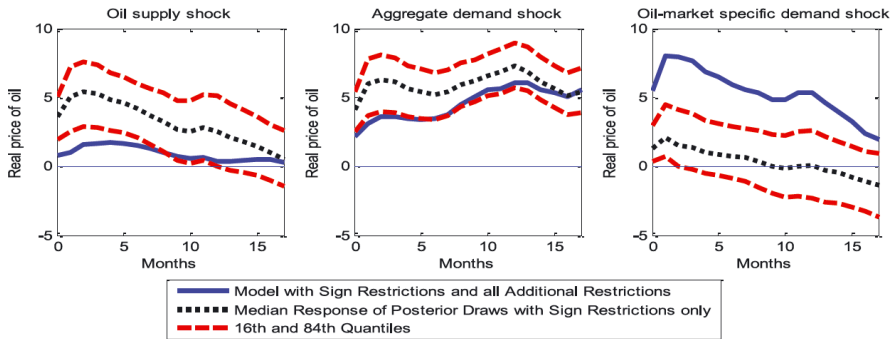
- Rubio-Ramírez, Waggoner & Zha (2010) show that, given the truncated Haar prior on Q , we can draw from the posterior of $(A(L), \Sigma, Q)$ as follows:
 - ① Draw from the normal-inverse-Wishart posterior of $(A(L), \Sigma)$ (this step is not affected by our choice of prior for Q).
 - ② Draw Q from (unrestricted) Haar measure.
 - ③ If $(A(L), \Sigma, Q)$ satisfy all restrictions, keep draw. Otherwise, discard Q draw and go back to step 2.
- Popular procedure in applied work due to computational simplicity.

Application: oil shocks

- There is nothing special about *sign* restrictions. Previous discussion applies also to *magnitude* restrictions, e.g.:

$$\Theta_{i,j,\ell} = B_{i,\bullet,\ell} C Q_{\bullet,j} \leq 0.5.$$

- Kilian & Murphy (2012) argue that adding magnitude restrictions can dramatically change Bayesian posterior distribution, given Haar prior.



Note: Blue line is posterior median with additional magnitude restrictions. Haar prior.

Sign restrictions: posterior convergence

- The prior on the identified reduced-form parameters $\nu = (A(L), \Sigma)$ gets dominated by the data for large T (Bernstein-von Mises Thm). Converges to a point mass at the true parameters $\nu = \nu_0$.
- But the conditional prior $\Pi(Q \mid \nu)$ on Q is never updated due to partial identification, as we have seen.
- How does the posterior behave for some general function $\theta = h(Q, \nu)$ of the parameters, such as an impulse response?
- Posterior for θ satisfies (under some conditions):

$$P(\theta \in \mathcal{B} \mid \mathcal{Y}_T) \rightarrow \Pi(\{Q: h(Q, \nu_0) \in \mathcal{B}\} \mid \nu_0) \quad \text{as } T \rightarrow \infty.$$

Depends on prior for Q even asymptotically.

Sign restrictions: posterior convergence (cont.)

$$P(\theta \in \mathcal{B} \mid \mathcal{Y}_T) \rightarrow \Pi(\{Q: h(Q, \nu_0) \in \mathcal{B}\} \mid \nu_0).$$

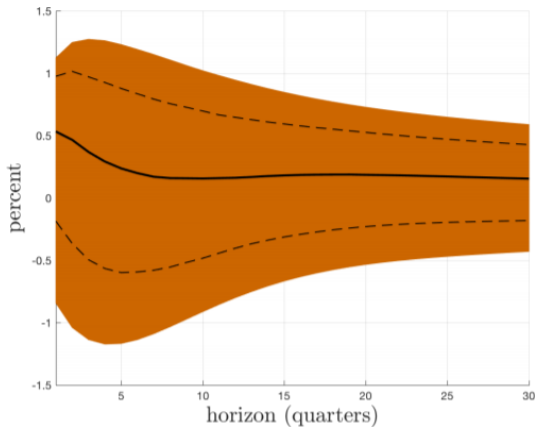
- The above distribution is concentrated on precisely those values of $\theta = h(Q, \nu)$ that lie in the identified set for θ , given $\nu = \nu_0$.
- Thus, the posterior for θ is some probability measure over the identified set for θ . Effectively, the prior for Q just provides a *weighting* of the parameters in the identified set.
 $\implies$ Any posterior credible set for θ lies strictly within the identified set, asymptotically (Moon & Schorfheide, 2012).
- Is this a problem?

Application: masquerading shocks

- Uhlig (2005) proposed an influential sign identification scheme for “identifying” monetary policy shocks: These shocks move interest rates and inflation in opposite directions.
- Wolf (2020) shows that if we apply this sign identification scheme to the Smets & Wouters (2007) DSGE model, the identified set is very wide (treating the ACF of the data as known, i.e., $T = \infty$).
- Although the monetary shock is the only standard shock that satisfies the sign restrictions, many other *linear combinations* of various shocks can “masquerade” as a monetary shock, in the sense of satisfying the sign restrictions.
- Why did the previous literature not realize that the identified set is so large? Because they just computed posterior credible bands imposing the (arbitrary) Haar prior. These are narrower than the identified set.

Application: masquerading shocks (cont.)

OUTPUT RESPONSE TO MONETARY SHOCK



Colored band: Identified set.

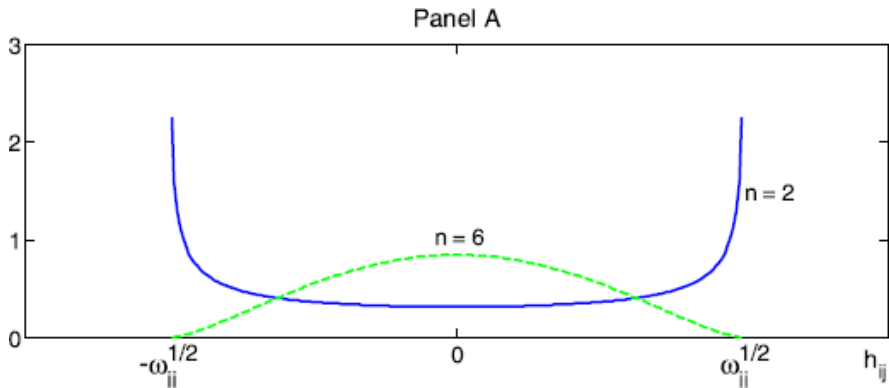
Dashed band: 68% credible band given Haar prior.

Source: Wolf (2020)

Sign restrictions: Haar prior

- We don't have any economic justification for the Haar prior, it's just numerically convenient. But is it possible that it may still imply a prior on impulse responses with “reasonable” properties?
- No. Baumeister & Hamilton (2015) show that the Haar prior implies a funky (conditional) prior on impulse responses.
- Specifically, consider the implied prior on the $(1, 1)$ element of the impact impulse response matrix $H = CQ$. For illustration, treat $C = I_n$ as known and draw Q from Haar measure.
- B&H show that the distribution of $H_{11} = Q_{11}$ depends dramatically on the number of variables n .
 - ▶ It changes from being unimodal to bimodal when n changes.
 - ▶ Density function can be bell-shaped, flat, or U-shaped.

Sign restrictions: Haar prior (cont.)



Explicitly subjective Bayesian inference

- General lesson: No prior is “agnostic” for all purposes. A “uniform” prior on some parameters may imply a very non-uniform prior for relevant parameter transformations.
- Baumeister & Hamilton (2015, 2018) propose to instead do subjective Bayesian inference where one is *explicit* about the prior placed on key economic parameters such as impulse responses or elasticities.
- They consider the SVAR model written in so-called **structural form**

$$\Xi y_t = \sum_{\ell=1}^p M_{\ell} y_{t-\ell} + \varepsilon_t, \quad \varepsilon_t \overset{i.i.d.}{\sim} N(0_{n \times 1}, \text{diag}(\sigma_1^2, \dots, \sigma_n^2)),$$

where the diagonal of Ξ is normalized to 1.

- They put a flexible prior on the short-run **elasticity matrix** Ξ and conjugate prior on everything else. The prior on Ξ is supposed to reflect scientific knowledge about relationships between variables, e.g., signs, likely magnitudes, etc. Inference: MCMC.

Explicitly subjective Bayesian inference (cont.)

- Another option is to put a prior directly on the impulse responses.
- Kocięcki (2010) shows how to transform a prior on impulse responses into a prior on VAR param's (using Jacobian of the transformation).
- Plagborg-Moller (2019): If we're in the business of putting priors on impulse responses, why not just work directly with the SVMA model?

$$y_t = \Theta(L)\varepsilon_t, \quad \Theta(L) = \sum_{\ell=0}^{\infty} \Theta_{\ell} L^{\ell}, \quad \varepsilon_t \overset{i.i.d.}{\sim} N(0_{n \times 1}, I_n).$$

- ▶ Requires specifying a high-dimensional prior, but at least the parameters are easily interpretable: they're just IRs. Fast MCMC algorithm.
- ▶ Don't have to assume invertibility.
- ▶ Some features of the prior will not get updated by the data due to the partial identification of the SVMA model.

Application: Bayesian inference in SVMA model

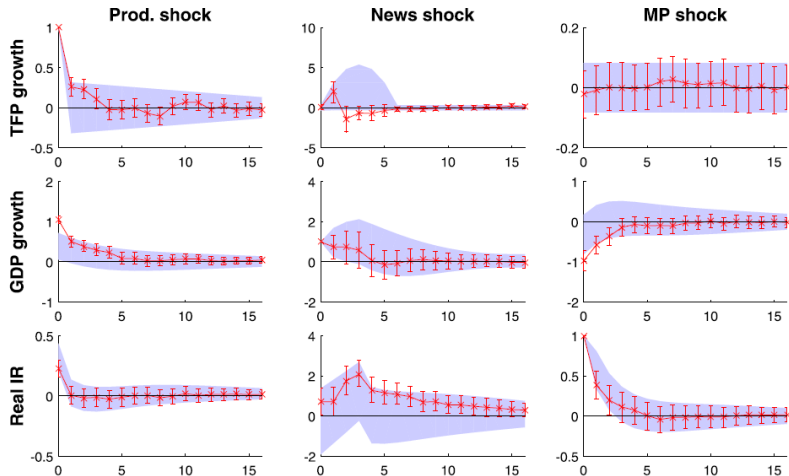


FIGURE 6. Summary of posterior IRF (θ) draws, news shock application. The plots show prior 90% confidence bands (shaded), posterior means (crosses), and posterior 5–95 percentile intervals (vertical bars).

Source: Plagborg-Møller (2019)

Sign restrictions: estimation of the identified set

- Instead of doing inference about the *parameters*, we could try to do inference on the entire *identified set*.
- We return again to the SVAR model with sign restrictions.
- Let $\mathcal{S}(\nu)$ be the identified set of a parameter of interest $\theta = h(Q, \nu)$. This set of course depends on the VAR parameters $\nu = (A(L), \Sigma)$.
- In most cases (with scalar parameter of interest), this set is convex, i.e., a connected interval with end-points obtained by the maximization/minimization mentioned earlier.
 - ▶ Sufficient conditions for convexity: Giacomini & Kitagawa (2021).
- Point estimate of the identified set:

$$\mathcal{S}(\hat{A}(L), \hat{\Sigma}).$$

Consistent if end-points are continuous fcts of VAR parameters.

Sign restrictions: Bayesian inference on the identified set

- To do inference, let us first adopt a Bayesian perspective.
- We seek a set $\hat{\mathcal{S}}$ that covers the true identified set with posterior probability $1 - \alpha$.
- Given posterior draws of $\nu = (A(L), \Sigma)$, this is conceptually easy:
 - ▶ Compute $\mathcal{S}(\nu)$ for every posterior draw ν .
 - ▶ For each possible interval $\hat{\mathcal{S}}$, check if $\hat{\mathcal{S}}$ contains at least $1 - \alpha$ of the posterior draws of the sets $\mathcal{S}(\nu)$.
 - ▶ Look for the shortest interval $\hat{\mathcal{S}}$ satisfying the above condition.
- Explicit computational algorithm: Giacomini & Kitagawa (2021).
 - ▶ Could compute $\mathcal{S}(\nu)$ via quadratic programming. Could also randomly sample Q from some distribution on $\mathbb{S}_n$, e.g., Haar, but – importantly – we do not use the quantiles of the saved Q draws, only the min+max.

Sign restrictions: robust Bayesian inference

- Giacomini & Kitagawa show that the previous procedure can be given a decision theoretic justification as a **robust Bayesian** credibility region for the parameter $\theta = h(Q, \nu)$.
- Fix some prior distribution $\Pi(\nu)$ for the reduced-form parameters (doesn't matter in large samples).
- Then the region $\hat{\mathcal{S}}$ from the previous slide satisfies

$$\inf_{\Pi(Q|\nu)} P(\theta \in \hat{\mathcal{S}} \mid \mathcal{Y}_T) \geq 1 - \alpha,$$

where the infimum is taken over *all* possible conditional priors for Q given ν that satisfy our sign restrictions.

- Guaranteed Bayesian credibility of at least $1 - \alpha$, *regardless* of the conditional prior on Q . Truly “agnostic”.

Sign restrictions: frequentist inference on the identified set

- There also exist methods for doing *frequentist* inference on the identified set (Imbens & Manski, 2004).
- Suppose the identified set is an interval $\mathcal{S} = [a, b]$.
- Suppose also we have managed to construct *one-sided* level- $(1 - \alpha/2)$ confidence intervals for the end-points:

$$\liminf_{T \rightarrow \infty} P(\hat{a} \leq a) \geq 1 - \alpha/2,$$

$$\liminf_{T \rightarrow \infty} P(\hat{b} \geq b) \geq 1 - \alpha/2.$$

- Then by Bonferroni's inequality, the interval $[\hat{a}, \hat{b}]$ is a valid level- $(1 - \alpha)$ confidence set for $\mathcal{S}$:

$$\begin{aligned} \limsup_{T \rightarrow \infty} P(\mathcal{S} \not\subset [\hat{a}, \hat{b}]) &= \limsup_{T \rightarrow \infty} P(a < \hat{a} \text{ or } b > \hat{b}) \\ &\leq \limsup_{T \rightarrow \infty} P(a < \hat{a}) + \limsup_{T \rightarrow \infty} P(b > \hat{b}) \leq \alpha/2 + \alpha/2 = \alpha. \end{aligned}$$

Sign restrictions: freq. inference on identif. set (cont.)

- If we can find asymptotically normal estimators of the bounds a and b of the identified set, then it would be easy to construct these one-sided confidence intervals.
- Unfortunately, in the case of sign-identified SVARs, the bounds are given as the max/min of an objective function subject to constraints.
 - ▶ The maximum/minimum is usually a continuous function of the underlying parameters of the problem (Theorem of the Maximum).
 - ▶ But the maximum/minimum may not be *continuously differentiable* in the parameters at points where there are multiple maxima/minima. So can't just apply delta method.
- However, Gafarov, Meier & Montiel Olea (2018) show that the bounds are *directionally differentiable*, and this is enough to apply a variant of the delta method and construct confidence intervals.

Sign restrictions: frequentist inference on parameters

- Any level- $(1 - \alpha)$ confidence set $\hat{\mathcal{S}}$ for the *identified set* $\mathcal{S}$ is also a (conservative) level- $(1 - \alpha)$ confidence set for the *true parameter* θ_0 :

$$P(\theta_0 \in \hat{\mathcal{S}}) \geq P(\mathcal{S} \subset \hat{\mathcal{S}}) \geq 1 - \alpha, \quad \text{since } \theta_0 \in \mathcal{S}.$$

- It is generally possible to construct a smaller set $\hat{\mathcal{S}}_\theta \subsetneq \hat{\mathcal{S}}$ with the property $P(\theta_0 \in \hat{\mathcal{S}}_\theta) \geq 1 - \alpha$ (Imbens & Manski, 2004; Stoye, 2009).
 - ▶ This is related to the literature on testing moment inequalities.
 - ▶ Granziera, Moon & Schorfheide (2018) develop such a procedure for sign-identified SVARs.
- In practice, the difference between $\hat{\mathcal{S}}$ and $\hat{\mathcal{S}}_\theta$ may not be large enough to justify the extra effort involved in computing the latter.